

Second order in homogeneous ODEs

Basic skills

Q1. a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 10y = 0$

auxiliary eq.
 $m^2 + 2m - 10 = 0$

$$(m+5)(m-2) = 0$$

$$m = -5, 2$$

2 real

$$y = Ae^{m_1x} + Be^{m_2x}$$

$$y = Ae^{-5x} + Be^{2x}$$

b) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

$$m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m = 3$$

1 real

$$y = Ae^{mx} + Bxe^{mx}$$

$$y = Ae^{3x} + Bxe^{3x}$$

c) $\frac{d^2y}{dx^2} + 9y = 0 \Rightarrow m^2 + 9 = 0 \Rightarrow m = \pm 3i$

2 complex

$$y = e^{rx}(A\cos 3x + B\sin 3x)$$

$$y = e^{0x}(A\cos 3x + B\sin 3x) \\ = A\cos 3x + B\sin 3x$$

Q2. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 14y = 0$

$$\Rightarrow m^2 + 5m - 14 = 0 \Rightarrow (m+7)(m-2) = 0$$

so $m = -7, 2$

so $y = Ae^{-7x} + Be^{2x}$

$y(0) = \frac{dy}{dx}(0) = 1$ so $A + B = 1$

$$-7A + 2B = 1$$

$$\frac{dy}{dx} = -7Ae^{-7x} + 2Be^{2x}$$

$$\Rightarrow A = 1 - B$$

$$\Rightarrow 9B - 7 = 1$$

so $B = \frac{8}{9}$ $A = \frac{1}{9}$ so $y = \frac{1}{9}(e^{-7x} + 8e^{2x})$

Q3. $y = 3\cos 2x \Rightarrow \frac{dy}{dx} = -6\sin 2x \Rightarrow \frac{d^2y}{dx^2} = -12\cos 2x$

so in $\frac{d^2y}{dx^2} + 7y = -12\cos 2x + 7 \times 3\cos 2x$
 $= 9\cos 2x$

so yes it does solve the ODE

Competency skills

Q4. $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = e^{2x}$

a) For homogeneous version

$$m^2 + m - 6 = 0 \Rightarrow (m+3)(m-2) = 0$$

$$\Rightarrow m = -3, 2 \quad \text{so } y_{CF} = Ae^{-3x} + Be^{2x}$$

b) $y = ae^{2x}$ $y' = 2ae^{2x}$ $y'' = 4ae^{2x}$

substituted into ODE gives

$$4ae^{2x} + 2ae^{2x} - 6 \times ae^{2x} = e^{2x}$$

$$\Rightarrow 0 = e^{2x} \quad \text{so no such } a$$

For $y = axe^{2x}$ $y' = 2axe^{2x} + ae^{2x}$
 $y'' = 4axe^{2x} + 4ae^{2x}$

so sub into ODE

$$4axe^{2x} + 4ae^{2x} + 2axe^{2x} + ae^{2x} - 6ae^{2x} = e^{2x}$$

$$\Rightarrow 5ae^{2x} = e^{2x} \Rightarrow a = \frac{1}{5}$$

$$\text{so } y_{PI} = \frac{1}{5}xe^{2x}$$

Hence general sol.

$$y = Ae^{-3x} + Be^{2x} + \frac{1}{5}xe^{2x}$$

c) Applying $y(0)=0$

$$\Rightarrow A+B=0$$

$$\frac{dy}{dx} = -3Ae^{-3x} + 2Be^{2x} + \frac{2}{5}xe^{2x} + \frac{1}{5}e^{2x}$$

Applying $\frac{dy}{dx}(0)=5$

$$\Rightarrow -3A + 2B + \frac{1}{5} = 5 \Rightarrow -3A + 2B = \frac{24}{5}$$

$$\text{so as } A = -B \Rightarrow 5B = \frac{24}{5} \quad B = \frac{24}{25}$$

$$\text{and } A = -\frac{24}{25}$$

not integral
d

so the particular solution is

$$y = \frac{1}{25} \left(24e^{2x} - 24e^{-3x} + 5xe^{2x} \right)$$

Q5. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 4\cos x$

For CF: $m^2 + 4m + 4 = 0 \Rightarrow (m+2)^2 = 0$
 $\Rightarrow m = -2$

So $y_{CF} = Ae^{-2x} + Bxe^{-2x}$

For PI: As $Q(x)$ is trig guess both

$$y = a \cos x + b \sin x$$

$$\frac{dy}{dx} = -a \sin x + b \cos x$$

$$\frac{d^2y}{dx^2} = -a \cos x - b \sin x$$

So in ODE

$$-a \cos x - b \sin x + 4(-a \sin x + b \cos x)$$

$$+ 4(a \cos x + b \sin x) = 4 \cos x$$

$$\Rightarrow (3a + 4b) \cos x + (3b - 4a) \sin x = 4 \cos x$$

$$\Rightarrow 3a + 4b = 4, \quad 3b - 4a = 0$$

$$\text{So } a = \frac{3}{4}b \text{ so } \frac{9}{4}b + 4b = 4 \Rightarrow b = \frac{16}{25}$$

and $a = \frac{12}{25}$

So $y_{PI} = \frac{4}{25} (3\cos x + 4\sin x)$

For general sol.

$y = Ae^{-2x} + Bxe^{-2x} + \frac{4}{25} (3\cos x + 4\sin x)$

Q6. $x \frac{d^2y}{dx^2} + 4xy = x^3 - x$, $y(0)=1$
 $y(2)=5$

$\frac{d^2y}{dx^2} + 4y = x^2 - 1$

CF: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$

So $y_{CF} = A\cos 2x + B\sin 2x$

PI: Guess polynomial

$y = ax^2 + bx + c$

Then $y'' = 2a$

So $2ax + 4(ax^2 + bx + c) = x^2 - 1$

SO $4a=1$, $2a+4b=0$, $4c=-1$

$$a = \frac{1}{4} \Rightarrow b = -\frac{1}{8} \text{ and } c = -\frac{1}{4}$$

SO $y_{PI} = \frac{1}{4}x^2 - \frac{1}{8}x - \frac{1}{4}$

SO general

$$y = A \cos 2x + B \sin 2x + \frac{1}{8}(2x^2 - x - 2)$$

APPLY $y(0) = 1 \Rightarrow A - \frac{2}{8} = 1 \Rightarrow A = \frac{5}{4}$

APPLY $y\left(\frac{\pi}{4}\right) = 5 \Rightarrow B + \frac{1}{8}\left(\frac{2\pi^2}{16} - \frac{\pi}{4} - 2\right) = 5$

$$\Rightarrow B = 5 - \frac{1}{8}\left(\frac{2\pi^2}{16} - \frac{\pi}{4} - 2\right)$$

SO $y = \frac{5}{4} \cos 2x + \left(5 - \frac{1}{8}\left(\frac{2\pi^2}{16} - \frac{\pi}{4} - 2\right)\right) \sin 2x$
 $+ \frac{1}{8}(2x^2 - x - 2)$

is the particular solution

Q7.

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = xe^{3x}$$

CF: $m^2 - 10m + 25 = 0$

so $y_{CF} = Ae^{3x} + Bxe^{3x}$

PI; could guess axe^{3x} but as
in CF pointless so try

$y = ax^2e^{3x}$ *multiply by x*

Then $y' = 2axe^{3x} + 3ax^2e^{3x}$

$$y'' = 2ae^{3x} + 10axe^{3x} + 10axe^{3x} + 25ax^2e^{3x}$$

Then $y'' - 10y' + 25y = xe^{3x}$
 $\Rightarrow 2ae^{3x} = xe^{3x}$ not possible

so need to try $y = ax^3e^{3x}$

then $y' = 3ax^2e^{3x} + 3ax^3e^{3x}$

$$y'' = 6axe^{3x} + 30ax^2e^{3x} + 25ax^3e^{3x}$$

Then $y'' - 10y' + 25y = xe^{5x}$

$$\Rightarrow 6ax e^{5x} = x e^{5x} \Rightarrow a = \frac{1}{6}$$

$$\text{so } y_{PI} = \frac{1}{6} x^3 e^{5x}$$

so general sol.

$$y = A e^{5x} + B x e^{5x} + \frac{1}{6} x^3 e^{5x}$$

Advanced skills

Q8. $3 \frac{dy}{dx} + 5y = 25 - 2 \sin 3x$

for $3 \frac{dy}{dx} + 5y = 0$ auxiliary $3m + 5 = 0$

$$\Rightarrow m = -\frac{5}{3} \text{ so } y_{CF} = A e^{-\frac{5}{3}x}$$

for particular try

$$y = a + b \cos 3x + c \sin 3x$$

$$\Rightarrow y' = -3b \sin 3x + 3c \cos 3x$$

Sub in to ODE

$$\Rightarrow 3(-3b \sin 3x + 3 \cos 3x) + 5(a + b \cos 3x + c \sin 3x) = 25 - 2 \sin 3x$$

$$\text{SO } 5a = 25, \quad 9c + 5b = 0, \quad 5c - 9b = -2$$

$$\text{SO } a = 5, \quad b = -\frac{9}{5}c$$

$$\Rightarrow 5c + \frac{81}{5}c = -2$$

$$\text{SO } \frac{106}{5}c = -2 \quad c = -\frac{10}{106}, \quad b = \frac{18}{106}$$

$$\text{SO } y_{PI} = 5 + \frac{18}{106} \cos 3x - \frac{10}{106} \sin 3x$$

SO general solution is

$$y = Ae^{-\frac{5}{3}x} + 5 + \frac{1}{53}(9 \cos 3x - 5 \sin 3x)$$

Q9.

$$\frac{d^2 c}{dt^2} + 50 \frac{dc}{dt} + 5000 c = 500 \cos 50t$$

a) auxiliary $m^2 + 50m + 5000 = 0$

$$m = \frac{-50 \pm \sqrt{50^2 - 4 \times 5000}}{2}$$

$$= -25 \pm 25\sqrt{7} i$$

$$\text{so } c_{CF} = e^{-25t} \left(A \cos(25\sqrt{7}t) + B \sin(25\sqrt{7}t) \right)$$

Particular integral

$$c = a \cos 50t + b \sin 50t$$

$$c' = -50a \sin 50t + 50b \cos 50t$$

$$c'' = -2500a \cos 50t - 2500b \sin 50t$$

so ODE becomes

$$2500 \left(-a \sin 50t + b \cos 50t + a \cos 50t + b \sin 50t \right) = 500 \cos 50t$$

$$\Rightarrow b - a = 0, \quad b + a = \frac{1}{5} \Rightarrow a = b = \frac{1}{10}$$

$$\text{So } C_{PI} = \frac{1}{10} (\cos 50t + \sin 50t)$$

for general sol

$$C = e^{-2s^2 t} (A \cos(25\sqrt{7}t) + B \sin(25\sqrt{7}t)) + \frac{1}{10} (\cos 50t + \sin 50t)$$

$$\text{Apply } C(0) = 0 \Rightarrow 0 = A + \frac{1}{10} \Rightarrow A = -\frac{1}{10}$$

$$\frac{dC}{dt} = e^{-2s^2 t} (-25\sqrt{7} A \sin(25\sqrt{7}t) + 25\sqrt{7} B \cos(25\sqrt{7}t))$$

$$- 2s e^{-2s^2 t} (A \cos(25\sqrt{7}t) + B \sin(25\sqrt{7}t)) + \frac{1}{10} (-50 \sin 50t + 50 \cos 50t)$$

$$\frac{dC}{dt}(0) = 0 \Rightarrow 25\sqrt{7} B - 2sA + s = 0$$

$$B = \frac{2sA - s}{25\sqrt{7}} = \frac{-2s - s}{25\sqrt{7}}$$

$$= -\frac{3}{10\sqrt{7}}$$

so $C = \frac{e^{-25t}}{10} \left(-\cos(25\sqrt{7}t) - \frac{3}{\sqrt{7}} \sin(25\sqrt{7}t) \right) + \frac{1}{10} (\cos 50t + \sin 50t)$

b) $\cos(25\sqrt{7}t) + \frac{3}{\sqrt{7}} \sin(25\sqrt{7}t) = R \cos(\theta + \phi)$

use $\theta = 25\sqrt{7}t$, then

$$R \cos \phi = 1, \quad R \sin \phi = \frac{3}{\sqrt{7}}$$

$$R^2 = 1^2 + \frac{9}{7} = \frac{16}{7} \quad R = \frac{4}{\sqrt{7}}$$

$$\tan \phi = \frac{3}{\sqrt{7}} \quad \phi = 48.6^\circ$$

And $\cos 50t + \sin 50t = R \cos(\theta + \phi)$

$$\theta = 50t, \quad R \cos \phi = 1, \quad R \sin \phi = 1$$

$$\Rightarrow R^2 = 2, \quad R = \sqrt{2} \quad \tan \phi = 1 \Rightarrow \phi = 45^\circ$$

Then $C = -\frac{e^{-25t}}{10} \frac{4}{\sqrt{7}} \cos(25\sqrt{7}t + 48.6^\circ) + \frac{\sqrt{2}}{10} \cos(50t + 45^\circ)$



$$c(t) \sim \frac{\sqrt{2}}{10} \cos(50t + 45^\circ) \text{ as } t \rightarrow \infty$$

No limit as oscillatory